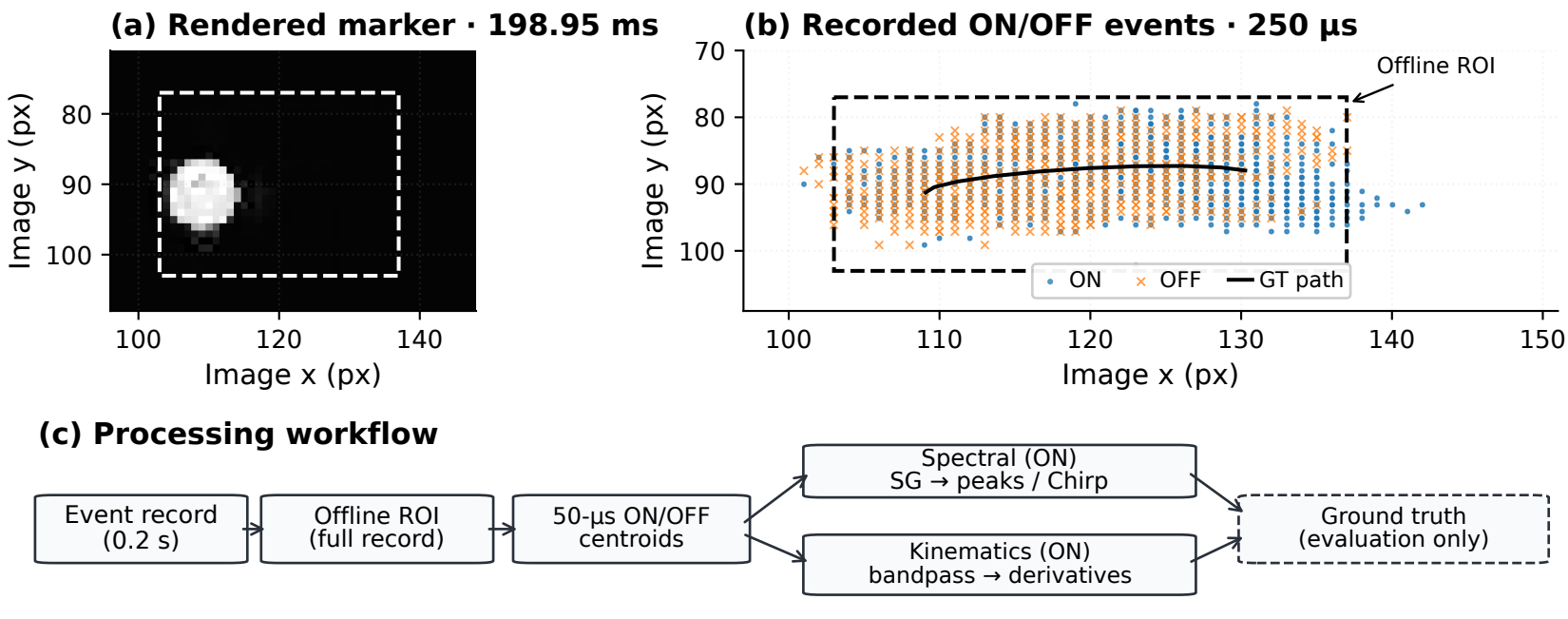
(a) Rendered marker · 198.95 ms
Image y (px)
80
90
100
100
120
140
Image x (px)
(b) Recorded ON/OFF events · 250 µs
Image y (px)
70
80
90
100
Offline ROI
ON
OFF
GT path
100
110
120
130
140
150
Image x (px)
(c) Processing workflow
Event record (0.2 s)
Offline ROI (full record)
50-µs ON/OFF centroids
Spectral (ON) SG → peaks / Chirp
Kinematics (ON) bandpass → derivatives
Ground truth (evaluation only)